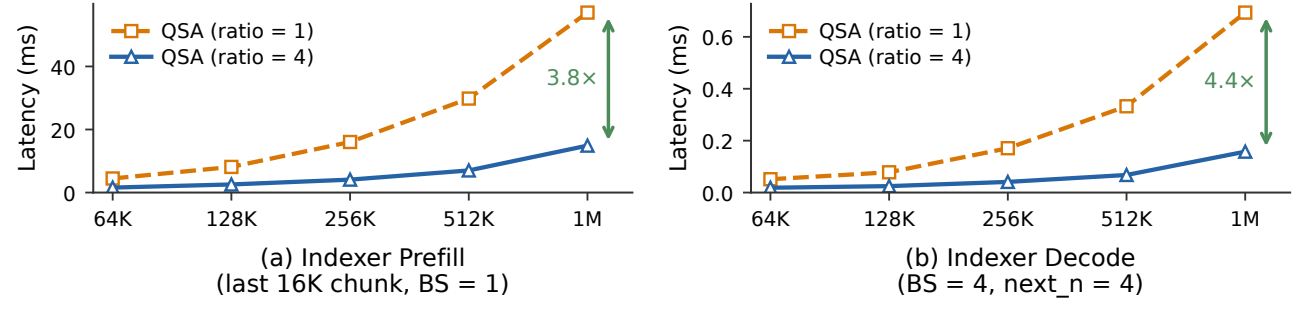
Latency (ms)
QSA (ratio = 1)
QSA (ratio = 4)
3.8×
0
20
40
64K
128K
256K
512K
1M
(a) Indexer Prefill
(last 16K chunk, BS = 1)
Latency (ms)
QSA (ratio = 1)
QSA (ratio = 4)
4.4×
0.0
0.2
0.4
0.6
64K
128K
256K
512K
1M
(b) Indexer Decode
(BS = 4, next_n = 4)
Latency (ms)
GQA
QSA
7.6×
0
100
200
64K
128K
256K
512K
1M
(c) Attention Prefill
(last 16K chunk, BS = 1)
Latency (ms)
GQA
QSA
4.9×
0.0
0.5
1.0
64K
128K
256K
512K
1M
(d) Attention Decode
(BS = 4, next_n = 4)